



ECEN 314 – Signals and Systems

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Module 18 - Analog to Digital Conversion - Sampling

Learning Objectives



- Understand and be able to develop/prove Nyquist's sampling theorem.
- For a given signal, be able to determine the necessary sampling rate for which perfect recovery is possible.
- Understand aliasing and how it will distort a signal.
- Understand how to represent bandpass signals at rates below the Nyquist rate and also what the lowest sampling rate is for bandpass signals.

Analog to Digital Conversion



As most signal processing is done digitally these days, it is necessary to convert signals that are inherently analog in nature to a digital form. There are several facets to this process:

Sampling – The process by which a signal is converted from continuous time to discrete time. The analog signal, $x(t)$, is represented by its samples $x[n] = x(nT_s)$.

Quantization - The process by which a signal is converted from continuous amplitude to discrete amplitude.

Encoding – The process by which each of a finite number of possible signal values are represented by a string of bits.

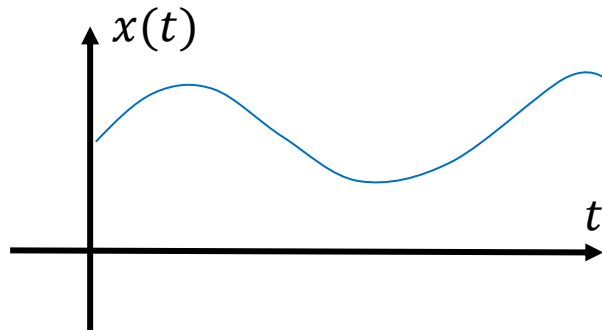
These slides will discuss issues associated with sampling. The other topics will be left for future courses (e.g., ECEN 303, 455).

Sampling

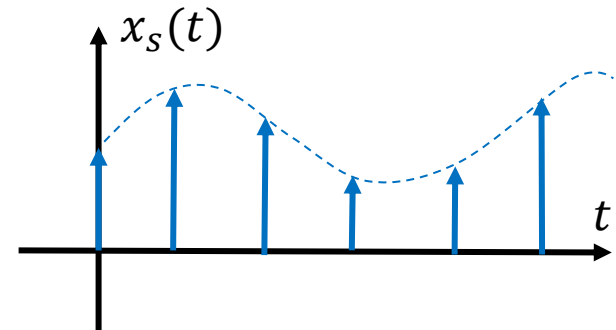


Suppose we are given samples of a signal spaced by T_s seconds (at a sampling frequency of $f_s = 1/T_s$). Is it possible to recover the original continuous time signal, $x(t)$, from its samples, $x[n] = x(nT_s)$? Under some surprisingly mild circumstances, the answer is “yes and perfectly (with no distortion).”

$$x(t) \longrightarrow \boxed{\text{Ideal sampler}} \longrightarrow x_s(t) = \sum_{k=-\infty}^{\infty} x(kT_s) \delta(t - kT_s)$$



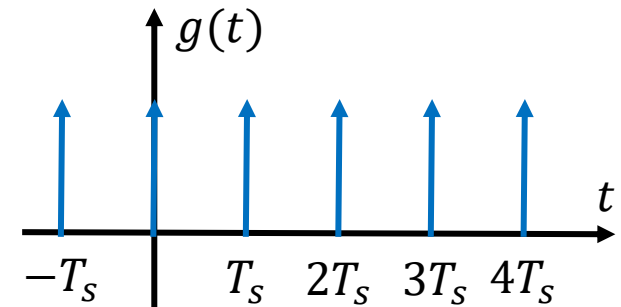
sampler



Define the ideal sampling function (impulse train),

$$g(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT_s),$$

So that $x_s(t) = x(t)g(t) \leftrightarrow X_s(f) = X(f) * G(f)$



Sampling

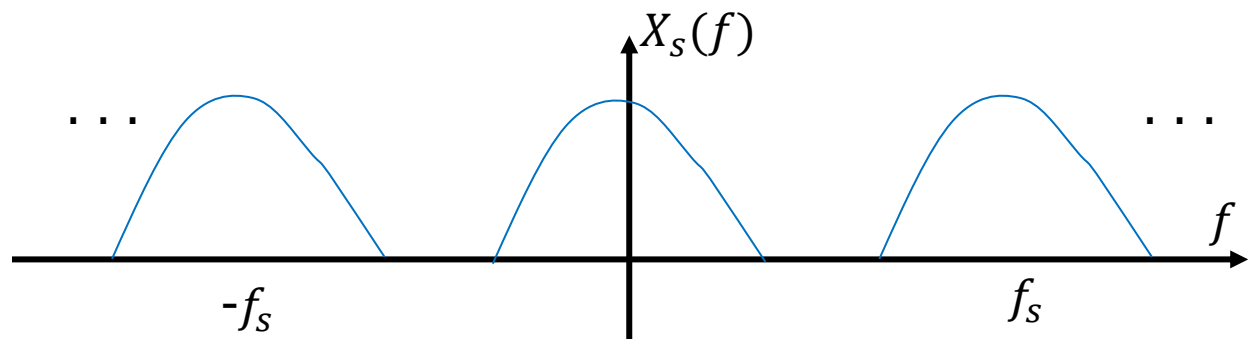
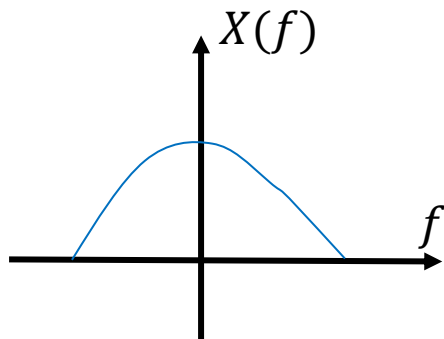


We found previously that the ideal sampling function has a spectrum given by

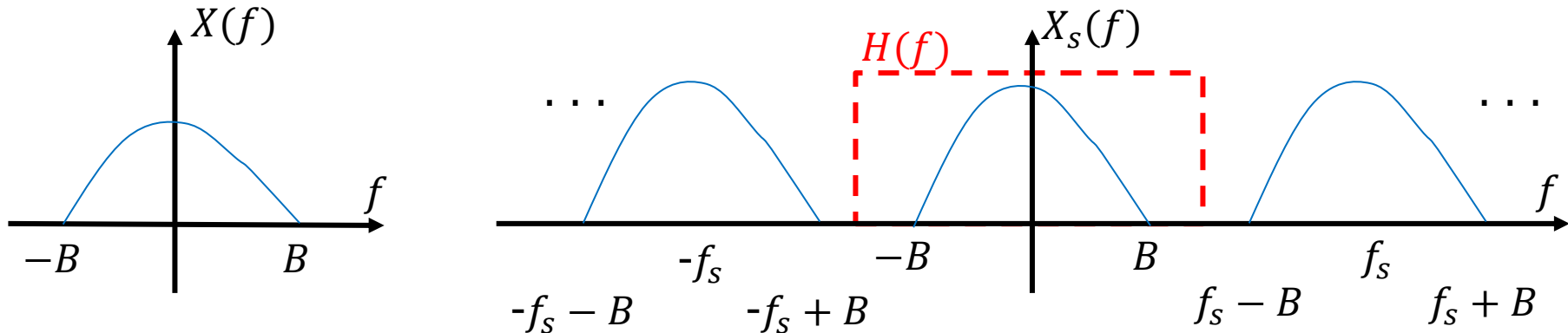
$$g(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT_s) \leftrightarrow G(f) = f_s \sum_{m=-\infty}^{\infty} \delta(f - mf_s).$$

This allows us to find the spectrum of the sampled signal

$$\begin{aligned} X_s(f) &= X(f) * G(f) = X(f) * [f_s \sum_{m=-\infty}^{\infty} \delta(f - mf_s)] \\ &= f_s \sum_{m=-\infty}^{\infty} X(f) * \delta(f - mf_s) = f_s \sum_{m=-\infty}^{\infty} X(f - mf_s) \end{aligned}$$



Sampling



Nyquist's Sampling Theorem: If $x(t)$ is bandlimited to B Hz (has an absolute BW of B Hz) then $x(t)$ can be perfectly recovered from it's samples provided that $f_s > 2B$. The sampling rate $f_s = 2B$ is known as the *Nyquist rate*.

In the time domain, the recovery is written in terms of a convolution

$$x(t) = x_s(t) * h(t) \text{ with } h(t) = \left(\frac{2f_c}{f_s}\right) \text{sinc}(2f_c t)$$

$$x(t) = \left(\frac{2f_c}{f_s}\right) \left(\sum_{k=-\infty}^{\infty} x(kT_s) \delta(t - kT_s)\right) * \text{sinc}(2f_c t)$$

$$= \left(\frac{2f_c}{f_s}\right) \sum_{k=-\infty}^{\infty} x(kT_s) \text{sinc}(2f_c(t - kT_s)).$$

Example

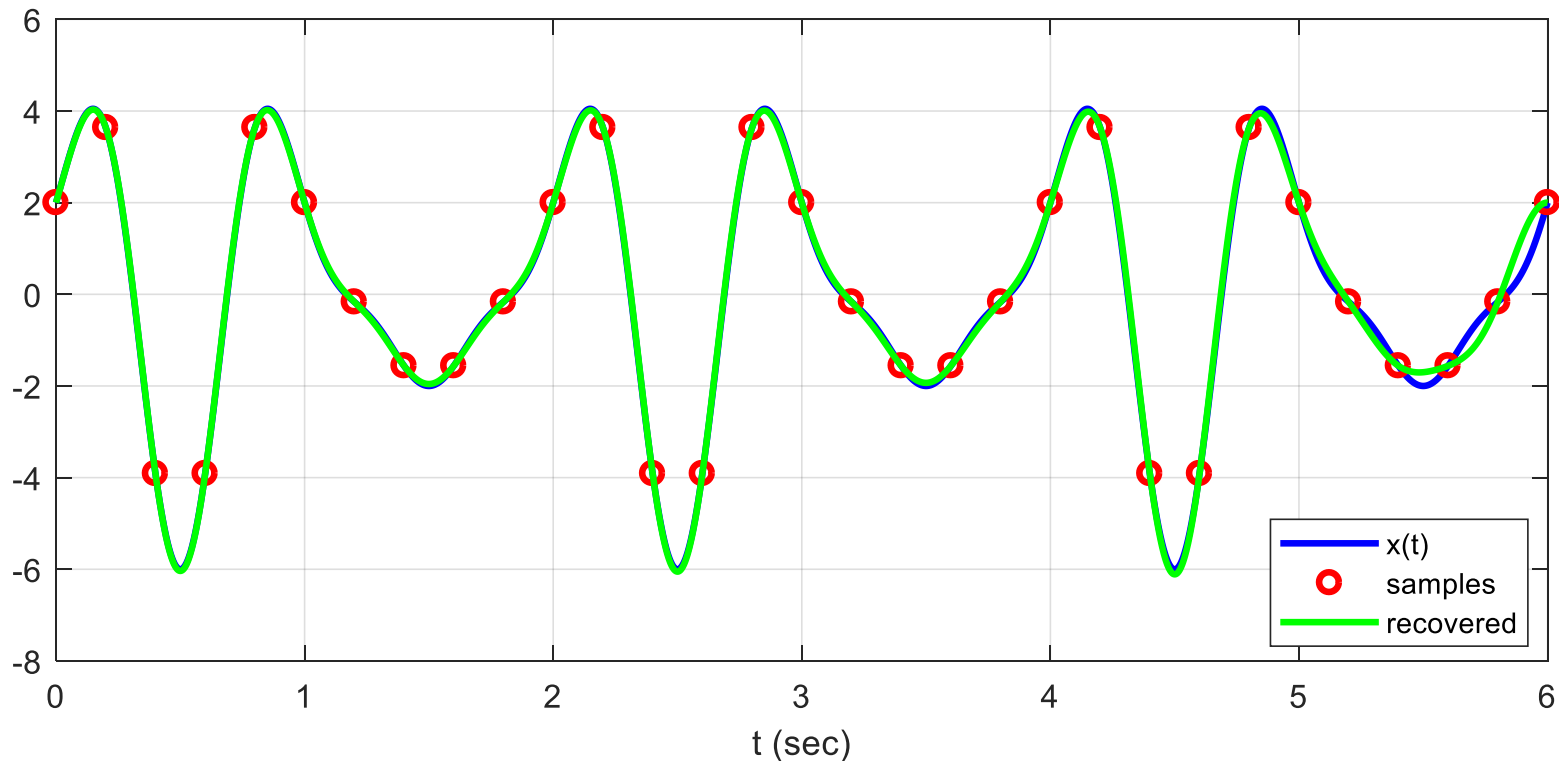
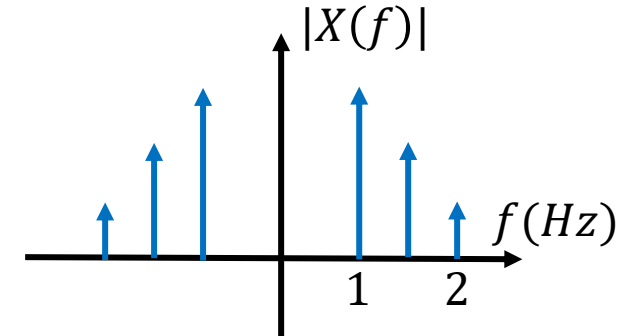


Suppose $x(t) = 3 \cos(2\pi t) + 2 \sin(3\pi t) - \cos(4\pi t)$.

$B = 2\text{Hz}$. $\rightarrow$ Nyquist rate $= 4\text{Hz}$.

Let's use $f_s = 5\text{Hz} \rightarrow T_s = 0.2\text{sec}$.

Using, $f_c = 2.5\text{Hz}$ the results below are obtained.



Practical Recovery Filter

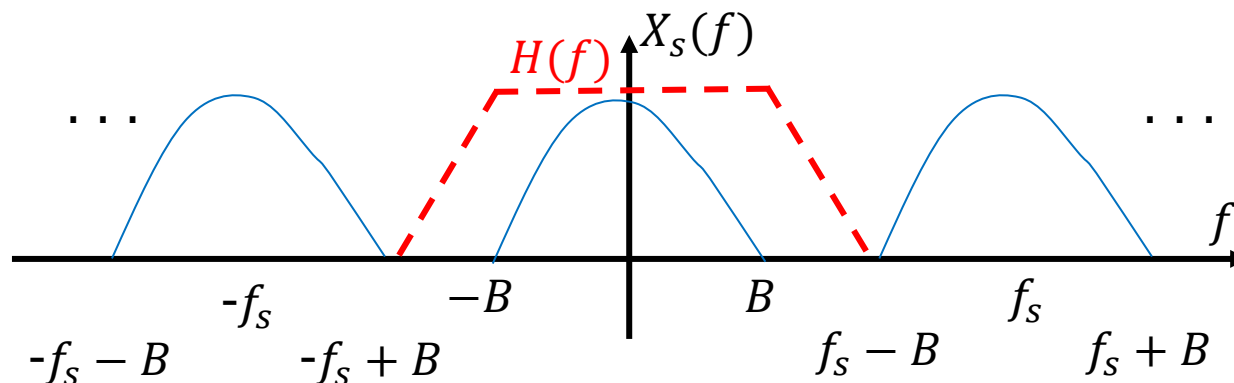


In practice, we must truncate the infinite series required to recover the signal from its samples.

$$x_r(t) = \sum_{k=-M}^M x(kT_s) \text{sinc}(2f_c(t - kT_s)).$$

Since the sinc signal dies off very slowly, we may need a large number of terms in the series to get an accurate recovery.

This can be improved by using a less than ideal LPF. The gradual rolloff in the frequency domain will result in a shorter duration of the impulse response in the time domain.



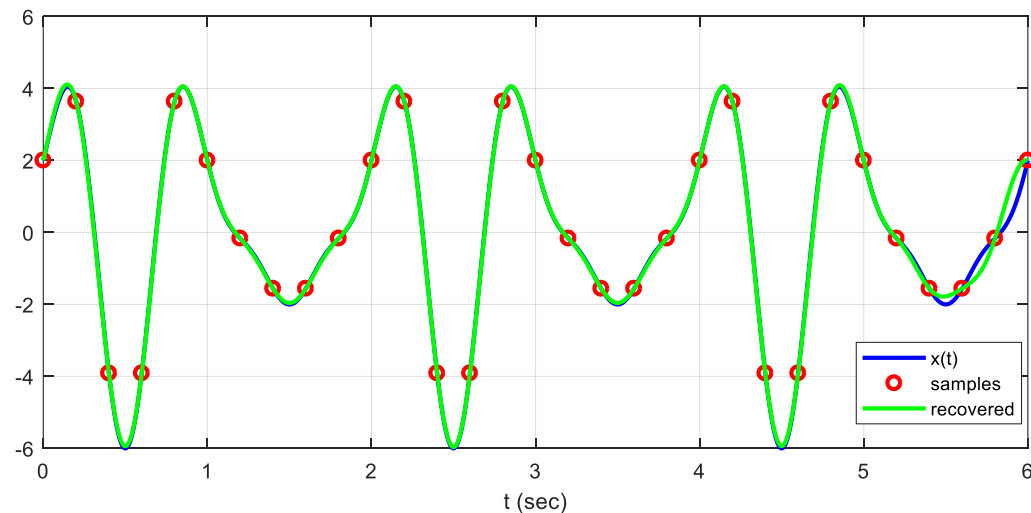
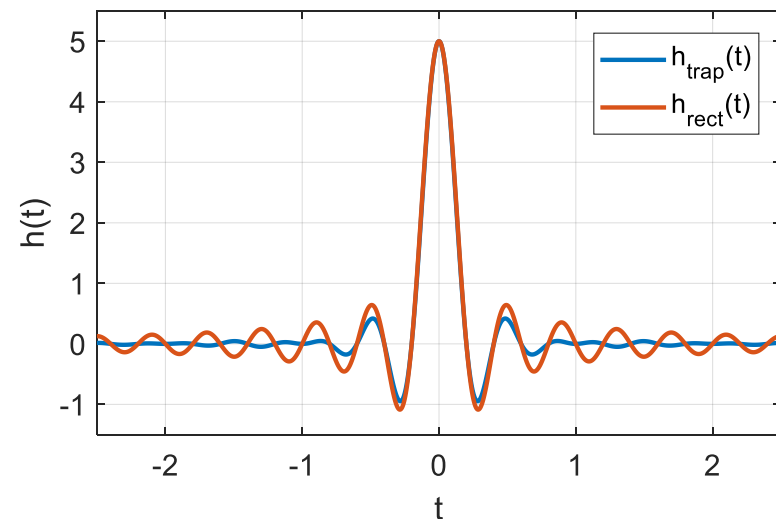
Practical Recovery Filter



These plots show the impulse response of the recovery filter for both the rectangular filter and the trapezoidal filter along with the recovered signal from the previous example.

$$h_{rect}(t) = f_c \text{sinc}(f_c t),$$

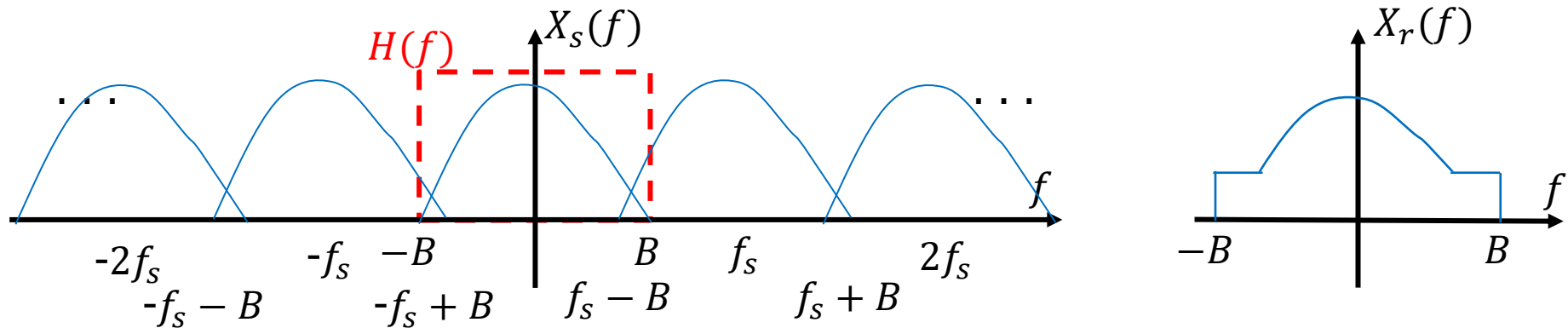
$$h_{trap}(t) = \frac{1}{f_s - 2B} \{ [(f_s - B) \text{sinc}((f_s - B)t)]^2 - [B \text{sinc}(Bt)]^2 \}.$$



Aliasing



If we try to sample at a rate below, the Nyquist rate, $f_s < 2B$, the recovered signal will be distorted.



This distortion of the high frequency components is known as *aliasing* or *spectral folding*.

- ❑ In general, the more we try to lower the sampling frequency, the worse the aliasing will become.
- ❑ There are some special cases where we can sample below the Nyquist rate without experiencing any aliasing, most notably signals that are bandpass in nature.

Aliasing

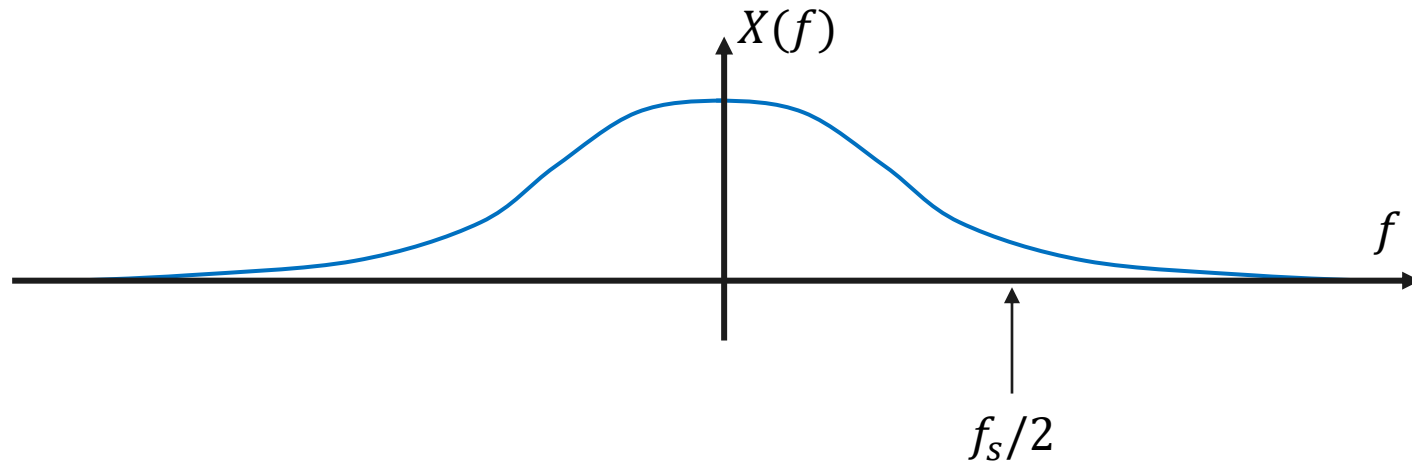


In practice, no signal is ever absolutely bandlimited. $\rightarrow B = \infty$.

$\rightarrow$ We will always be sampling below the Nyquist rate since strictly speaking $2B = \infty$.

$\rightarrow$ There will always be some degree of aliasing.

As engineers, we manage the aliasing to keep it at an acceptable level.



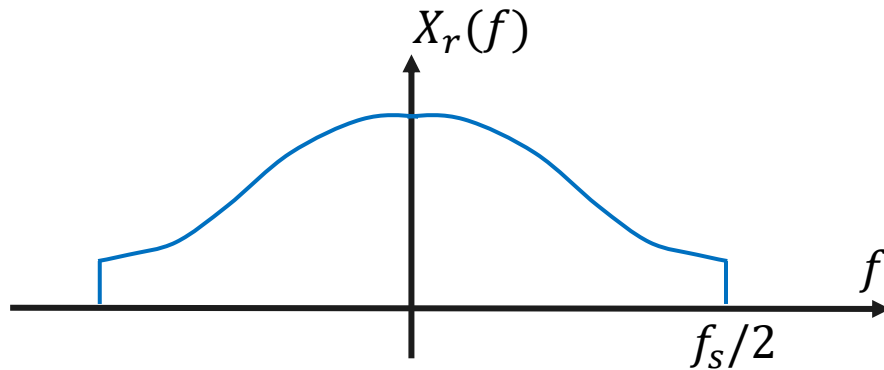
Should we:

- Just sample at whatever rate we choose to use and let a little bit of aliasing happen, or
- Pre-filter (distort) our signal before sampling to remove any frequency content above $f_s/2$ so that there will be no aliasing?

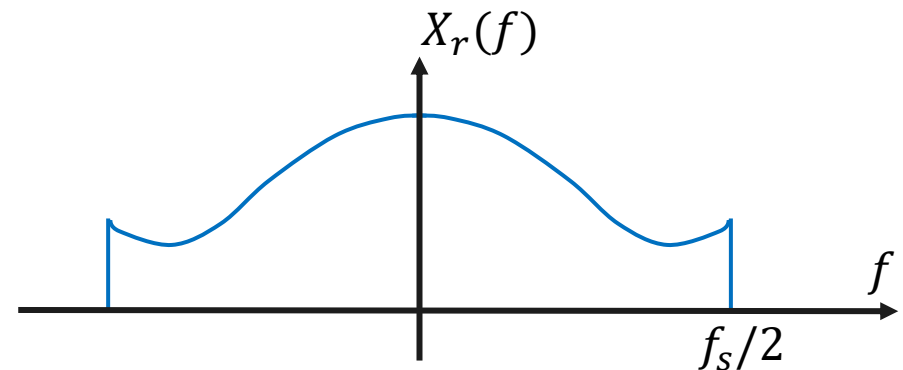
Aliasing



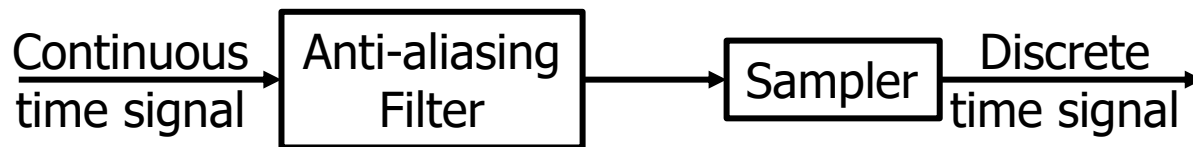
The answer is that we will end up with less distortion if we remove any frequency content above $f_s/2$ from our signal before sampling.



Recovered Spectrum with pre-filtering



Recovered Spectrum without pre-filtering



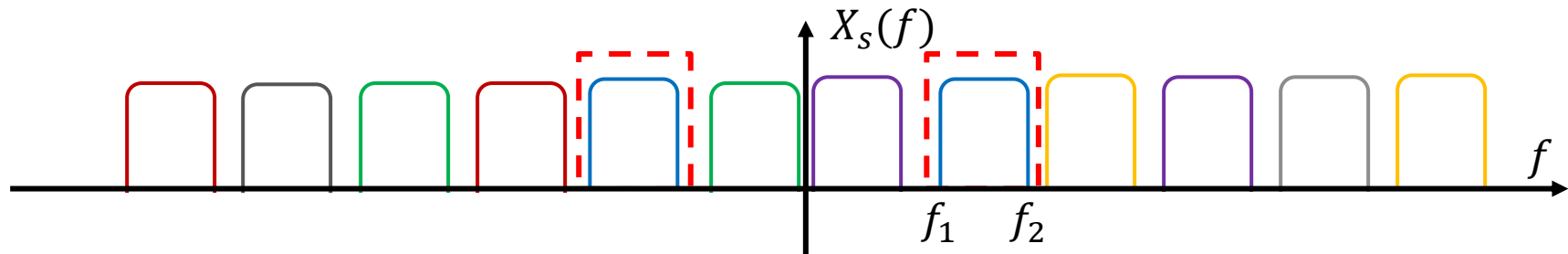
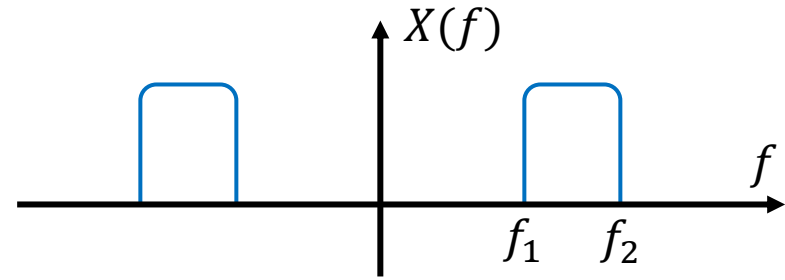
The pre-filter is known as an anti-aliasing filter and since it occurs prior to sampling, it must be implemented as an analog filter.

Once the signal is converted to discrete time, then any remaining signal processing we might want to do can be implemented digitally (in software).

BandPass Sampling



If a signal's spectrum is bandpass in nature, there may be some sampling rates below the Nyquist rate at which it is possible to perfectly recover the signal.



In this case, a BPF is needed to recover the original signal from its sampled version.

The allowable sampling frequencies for which perfect recovery is possible is a rather convoluted function of f_1 and f_2 . It turns out the minimum possible sampling rate is

$$f_{s,min} = \left\lceil \frac{f_2}{\frac{f_2}{f_2 - f_1}} \right\rceil.$$

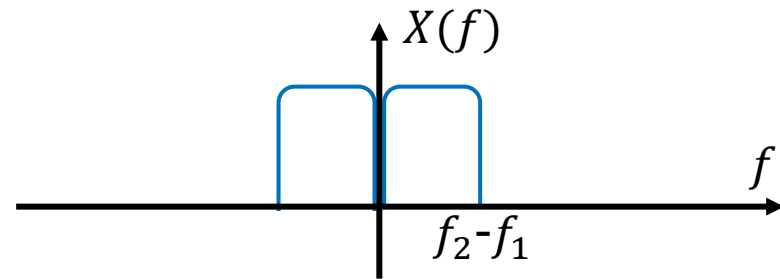
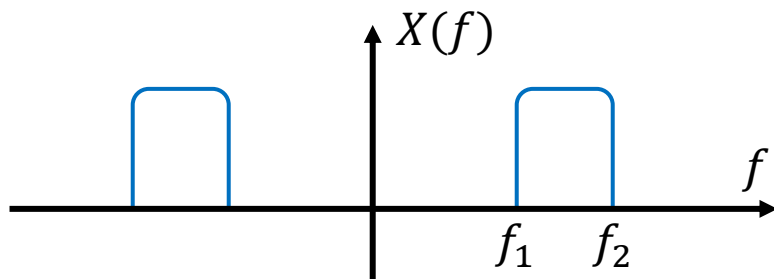
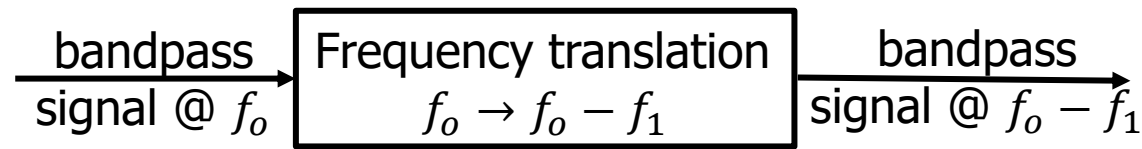
BandPass Sampling



Given the previous formula, the lowest possible sampling rate we can hope to achieve occurs when $f_2 = m(f_2 - f_1)$ (for an integer value of m). In that case,

$$f_{s,min} = \frac{2f_2}{m} = 2(f_2 - f_1) \rightarrow \text{twice the bandwidth.}$$

Rather than sampling the bandpass signal directly, the same result can always be achieved if we first do some processing of the signal to translate the signal to a lower frequency band.

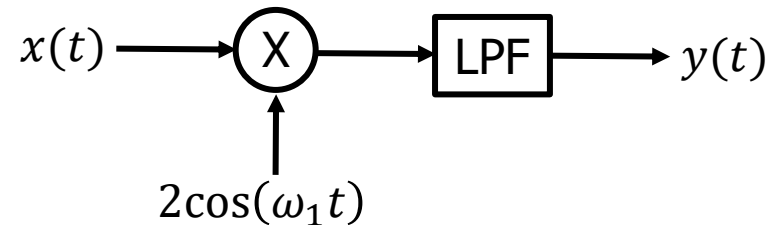


Once the signal is translated in frequency, its largest frequency component is now $f_2 - f_1$ and it can be sampled at any rate $> 2(f_2 - f_1)$.

BandPass Sampling



Frequency translation can be implemented using a system as shown in the block diagram.



The input to the lowpass filter is

$$2x(t) \cos(\omega_1 t) = x(t)[e^{j\omega_1 t} + e^{-j\omega_1 t}].$$

In the frequency domain this signal's spectrum is

$$\begin{aligned} X(f) * [\delta(f + f_1) + \delta(f - f_1)] \\ = X(f + f_1) + X(f - f_1). \end{aligned}$$

Therefore, the spectrum of $x(t)$ has been shifted up by f_1 and shifted down by f_1 . The lowpass filter allows the downshifted version to pass and the upshifted version is blocked.

→ $y(t)$ is just $x(t)$ downshifted by f_1 Hz.

Notes:

- This system could function as an upconverter if the LPF is replaced with an HPF.
- We will discuss how to practically implement this sort of block diagram a bit later in the course.